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Adiabatic and Nonadiabatic Effects in Vibration–Rotational Spectra of Diatomic Molecules

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For the vibration–rotational motion of a diatomic molecule, various forms of an effective Hamiltonian which includes the corrections of the Born–Oppenheimer approximation in the form of radial functions are reviewed. A procedure to fit vibration–rotational and pure rotational transitions is proposed which is insensitive to the specific representation of the Born–Oppenheimer corrections but is capable of extracting the mass-independent Born–Oppenheimer potential from the experimental data. This method is applied to LiH successfully. The truncation of the power expansion of the Born–Oppenheimer potential introduces ambiguity depending on the mathematical form of the expansion coordinate. We reconsider the question of separability of the adiabatic and nonadiabatic corrections which can be derived from the atomic mass dependence of the molecular energy levels. We demonstrate that by means of a measured molecular rotational g-factor g_J, the adiabatic and nonadiabatic terms of the rotational energy become separable in lowest order. The possibility to estimate the lowest order for the correction function of the vibrational kinetic energy operator is briefly discussed. The inclusion of this contribution is important to allow a comparison of the various existing models for analysis of vibration–rotational spectra of great precision. In the case of LiH it is shown that the higher order mass corrections are dominated by the contributions of the rotating and vibrating electric dipole moment. © 1994 Academic Press, Inc.

INTRODUCTION

From applications of the highly precise methods of both microwave spectroscopy, and infrared spectroscopy using interferometers with Fourier transforms or diode lasers, many data have been published on simple diatomic molecules such as LiH (*1–3*). Such data were used to test the quality of the Born–Oppenheimer approximation and to derive information on the adiabatic and nonadiabatic corrections to that approximation. For LiH, several analyses already exist, including some recent papers (*3–5*). There is some controversy about the methods used and their general application. Because Maki *et al.* (*3*) failed to obtain a convincing fit of their data, they questioned the validity of existing theory (for example see Ref. *6*) for light molecules like hydrides. The nonadiabatic effects might be large for highly rotationally excited states for which the perturbation calculation of the interaction between electrons and nuclei could be inadequate up to second order.

Both Ogilvie (*4*) and Coxon (*5*) report on a successful analysis of full experimental data but their procedures are incomplete because they omit the contribution of the rotational g-factor g_J, which is expected from an early discussion of the Born–Oppenheimer approximation by van Vleck (*7*) and given later in more details by several

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authors (e.g., see Ref. *9*). This contribution must be hidden in some of the many parameters used to fit the data (*4, 5*).

We use LiH only to demonstrate our point of view, but LiH is especially suitable for the general discussion of problems concerning the quality of the Born–Oppenheimer approximation and necessary corrections because precise data of four isotopic variants exist, and because the correction terms are expected to be relatively large for such a light molecule.

We first review briefly the status of existing theoretical models of the above topic and their applications to spectral analysis. Then we introduce a more general fitting procedure which can be taken in part as a combination of the well established method from the theory developed by Watson (*6*) and the one used by Ogilvie (*4*). Results from fits with different methods are compared to show what part of the derived molecular parameters are model independent and what are more or less model dependent and therefore not really adequate for physical interpretation. In the final section we include further information from other sources to separate as much as possible the adiabatic and nonadiabatic correction terms from each other.

REVIEW OF THE CORRECTIONS TO THE BORN-OPPENHEIMER APPROXIMATION

The vibration–rotational energy of a diatomic molecule that follows rigorously the Born–Oppenheimer approximation can be calculated from the Schrödinger equation with the Hamiltonian H including the mass-independent potential energy $V(R)$ of a state $\Omega = 0$,

$$H = P_R^2/2\mu + V(R) + B_e(R_e/R)^2 J(J+1), \tag{1}$$

in which μ is the atomic reduced mass of the molecule and B_e the rotational constant related to the equilibrium internuclear distance R_e. Using the expansion of $V(R)$ as a series in

$$x = (R - R_e)/R_e \qquad \text{giving } V(R) = a_0 x^2 (1 + \sum_j a_j x^j) \tag{2}$$

with positive powers about $x = 0$, Dunham (*8*) showed that the energy levels follow a simple power expansion with the vibrational (v) and rotational (J) quantum numbers. The total energy expression is written in the form (with U_{kl} in units of wavenumbers $\times u^{k/2+l}$)

$$E(v, J)/hc = \sum_k \sum_l U_{kl} \mu^{-k/2-l} (v + 1/2)^k J^l (J+1)^l; \tag{3}$$

here the molecular parameters U_{kl} are independent of mass (at least to a high degree of approximation; for more details see Eq. (7) later) and are simple functions (compare Ref. (*8*) and citations in Refs. (*3, 12*)) of the potential-energy coefficients a_j defined by Eq. (2).

The perturbation calculation by Herman and Asgharian (*9*) for the deviations of the Born–Oppenheimer approximation resulted in the effective Hamiltonian of the vibration–rotational motion including the electronic nuclear interaction up to second order,

$$H = 1/(2\mu) P_R (1 + \beta(R)) P_R + V(R) + V_{cor}(R) + B_e(R_e/R)^2 J(J+1)(1 + \alpha(R)), \tag{4}$$

in which α, β, and V_{cor} are contributions to correct the Born–Oppenheimer approximation and result from matrix elements between electronic states or from expectation values, and are functions of the internuclear distance parameter R and the atomic masses involved. Therefore, the effective form is customarily written as

$$\alpha(R) = m_e(\alpha^A(R)/M_A + \alpha^B(R)/M_B)$$

$$\beta(R) = m_e(\beta^A(R)/M_A + \beta^B(R)/M_B)$$

$$V_{cor}(R) = m_e(V_A(R)/M_A + V_B(R)/M_B), \tag{5}$$

in which m_e is the electron mass and M_A and M_B are the atomic masses for a molecule AB. The correction functions $\alpha^{A,B}$, $\beta^{A,B}$, and $V_{A,B}$ differ for the two atoms A and B (if their atomic numbers differ) but are believed to be independent of nuclear mass for an atom of a given atomic number. This expression is expected to be correct within the perturbation order derived in Refs. (*9, 6*).

The effective Hamiltonian according to Eq. (4) has not yet been used to fit molecular spectra. Simplifications of two kinds are applied: either neglecting α and β and using only V_{cor}, which is then referred to as the adiabatic correction of the potential energy (called simplification 1 hereafter), or neglecting β, to render the Schrödinger equation in the conventional form with an effective potential composed of V, V_{cor}, and α (simplification 2). Following empirical precedent (*10*), Watson (*6*) showed for simplifications 1 and 2 that the eigenenergies of such a molecule can be written in a power series similar to that of Eq. (3),

$$E(v, J)/hc = \sum_k \sum_l U_{kl}\mu^{-k/2-l}(1 + m_e\Delta^A_{kl}/M_A + m_e\Delta^B_{kl}/M_B)(v + 1/2)^k J^l(J + 1)^l. \tag{6}$$

Watson generated no direct relations between the functions of Eq. (5) and the quantities Δ in Eq. (6). In this respect Eq. (6) is an effective representation of the vibration–rotational energies and is commonly applied to fit directly observed spectra. Ogilvie (*11*) made first attempts to relate the Δ to a radial function as $V_{cor}(R)$ according to the simplification 1. This treatment was extended (*4*) to include the function α of Eq. (4); recently, Fernandez and Ogilvie (*12*) gave a full perturbation treatment to relate all three functions, namely, α, β, and V_{cor}, to the Δ, in which these functions are given as a Taylor expansion of an appropriate normalized internuclear distance, for example with that of Eq. (2). Then similarly to the relations between U_{kl} and a_j (see, for example, Ref. *8*) there exists an equation for each Δ_{kl} which relates Δ to the expansion coefficients of V, α, β, and V_{cor} (*12*). Another important result of that paper (*12*) is the demonstration that for the full effective Hamiltonian of Eq. (4) the energies can be represented in the form of Eq. (6); this property might be taken as the foundation of many successful applications of Eq. (6) already in the literature not knowing the full range of validity.

Principally, Eq. (6) can hence be rewritten introducing completely for the U_{kl}, Δ^A_{kl}, and Δ^B_{kl} the various expansion coefficients of the functions V, α, β, and V_{cor};

i.e., V as a function of $a_0, a_1, a_2, \ldots$ and

α as a function of $\alpha_0, \alpha_1, \alpha_2, \ldots$ with $\alpha = \sum_j \alpha_j x^j$

and similarly for β and V_{cor}. (2′)

Fitting such representation to experimental observations would give directly the expansion coefficients of $V(R)$, α, β, and V_{cor}, if in principle they would be independently determinable and if in practice their correlations resulting from experimental observations would be not too severe. According to this approach Ogilvie (*4*) treated data of the molecule LiH with a nonlinear least-squares fit using simplification 2. In the following sections we will come back to the question if this method can be extended to the full form of Eq. (4).

For practical applications one has to truncate Eq. (6) at some point; this truncation may affect the model of the molecular energy levels in two ways, either directly according to the limits of k and l of the coefficients U_{kl}, i.e., to the powers of v and J, or indirectly according to the number of expansion coefficients a_j determinable by U_{kl} or used in the fit, which means to the powers of the expansion coordinate. In the latter case an indefinite number of U_{kl} should be used for a uniquely defined mathematical procedure for calculating the energies, and in the first case an indefinite number of a_j would be defined by $U_{kl} = 0$ for k and l beyond these limits mentioned above. Whether this would be possible mathematically or physically in a unique way will be discussed in the application section for LiH. Because energies of states specified by quantum numbers are measured by the spectroscopist the first mentioned truncation is generally used, although this practice is not invariably specified in the papers.

To overcome the problem of truncation, Coxon (*5*) used the numerical integration of the Schrödinger equation with a Hamiltonian according to simplification 2 to calculate the energy levels introducing an analytic representation of the potential function by a finite polynomial of a chosen expansion coordinate for example according to Eq. (2). The fit of the observed spectrum was performed applying the inverse perturbation approach IPA (*13*). Coxon argued (*5*) that the algebraic representation of Eq. (6) by the functions of Eq. (5), because it lacks a quantum-mechanical basis, cannot be used to predict new molecular transitions and that the reliability of the derived potential is not established. We consider that this lack has been remedied by the work in Ref. (*12*) and that the applications of the various approaches to the same data set of a molecule should show the limitations and benefits.

AN ITERATIVE FITTING PROCEDURE

The sets of energy levels given by the general form of Eq. (6) can represent Hamiltonians different from those given by Eq. (4), because it is shown only that the eigenspectrum of the operator in Eq. (4) can be represented by Eq. (6) but not that any spectrum of the form of Eq. (6) must have a Hamiltonian of the form of Eq. (4). Therefore, fitting molecular spectra to the formula (6) should use constraints to restrict to Hamiltonians of Eq. (4) which are then the source for physical interpretation.

For such constrains we will use interrelations between the U_{kl} but not between the Δ_{kl} which are only poorly determinable in most cases and therefore will not serve this purpose sufficiently well. From the algebraic approach (e.g., see Refs. (*8, 12*)) we know that each U_{kl} is a function of all coefficients a_j determining $V(R)$, but these can be ordered according to the perturbation degree,

$$U_{kl} = U_{kl}^{(1)} + U_{kl}^{(2)} 1/\mu + U_{kl}^{(3)} 1/\mu^2 + \cdots, \tag{7}$$

where the magnitude of each contribution decreases by the ratio $(U_{01}/U_{10})^2$ which is in most cases of the order of 10^{-5} to 10^{-6}. The factor by the reduced mass μ was extracted to keep all $U_{kl}^{(i)}$ mass independent. The calculations of Ref. (*8*) or Ref. (*12*) prove that each order of a $U_{kl}^{(i)}$ depends only on a finite and generally small number of a_j beginning with a_1. In the case of $U_{kl}^{(1)}$ the limit is precisely

$$j \leqslant 2k + l - 2. \tag{8}$$

Therefore, a selected coefficient $U_{kl}^{(1)}$ can be calculated with the help of several $U_{mn}^{(1)}$, where m and/or n is smaller than k and/or l. Such equations are called interrelations between the $U_{kl}^{(i)}$, the by far best known one is certainly the Kratzer relation $U_{02}^{(1)} = -4U_{01}^{(1)3}/U_{10}^{(1)2}$. Other relations can be calculated from the tables published by Dunham (*8*) and other authors (see citations in Refs. (*3, 12*)). Invoking such interrelations during the fit, we will search for a representation of the observed molecular levels by Eq. (6) which relates to a Hamiltonian of the form of Eq. (4).

In order to use linear least-squares methods, which are much more stable for large calculations, we consequently set up an iterative fit of the observed transition frequencies to Eq. (6) for the energy levels. After the first iteration with no constraints by interrelations all possible potential coefficients are derived from the two series of $U_{k0}^{(1)}$ and $U_{k1}^{(1)}$, which represent the lowest order for both the vibrational and rotational energies, respectively. Using Eq. (8) one obtains easily, if $U_{k0}^{(1)}$ and $U_{k-1,1}^{(1)}$ are the highest determined parameters, the potential coefficients a_j up to $j_{\max} = 2k - 2$ are derivable and in the case $U_{k0}^{(1)}$ and $U_{k1}^{(1)} j_{\max} = 2k - 1$.

Then the centrifugal-distortion coefficients $U_{kl}^{(1)}$ with $l > 1$ (what $U_{kl}^{(1)}$ with $l > 1$ have to be included results again from Eq. (8): $l_{\max} = j_{\max} + 2$ and $0 \leqslant k \leqslant \frac{1}{2}(j_{\max} - l) + 1$ with $1 < l \leqslant l_{\max}$) and the mechanical corrections to the $U_{k0}^{(1)}$ and $U_{k1}^{(1)}$—the so-called Dunham corrections $U_{kl}^{(2)}$, etc.—are calculated and a second iteration is started. With these corrections the U's in Eq. (3) and (6) have to be replaced by

$$U_{kl} \rightarrow U_{kl}^{(1)} + U_{kl}^{(2)} 1/\mu + \cdots \qquad (3') \text{ and } (6'),$$

to obtain the correct mass dependence and to remove contributions by $U_{kl}^{(i)} i > 1$ from the Δ_{kl}. The latter will now only contain contributions of the functions V_{cor}, α, β, according to Eq. (5). One should keep in mind that by this procedure our Δ parameters differ slightly from the common practice in the literature, where the Δ parameters always contain contributions by $U_{kl}^{(i)}$ with $i > 1$ to simplify the fitting routine. The iteration loop terminates when the changes in the potential-energy coefficients a_j of $V(R)$ become insignificant between successive iterations.

An important advantage of this procedure is that no starting values of the molecular parameters are necessary unlike the need in nonlinear fitting (*4, 5*); results of fits of the latter type are susceptible to dependence on initial values of the parameters. Additionally, for the new procedure a natural choice of Dunham parameters U_{kl} is automatically incorporated because to calculate one further potential-energy coefficient a_j the necessity of fitting U_{k0} or U_{k1} alternates for increasing k according to Eq. (8).

APPLICATION TO LiH

For the calculations to be discussed in this section the atomic masses were taken from Ref. (*14*); nuclear masses are unnecessary if one uses the appropriate reinterpretation of the α and β functions as introduced in Ref. (*9*). The frequencies and wavenumbers of transitions of the isotopic variants of LiH were taken from Refs. (*1, 3*); the millimeterwave wavenumbers were assigned uncertainties 1.7×10^{-6} cm^{-1}, the far-infrared data 0.0006 cm^{-1} and the remaining infrared data 0.0008 cm^{-1}. The data of Ref. (*2*) with their stated uncertainties 0.001 cm^{-1} show systematic shifts of about +0.001 cm^{-1} compared to the transitions given in Ref. (*3*), which are slightly more precise and are significantly more extended than those of Ref. (*2*). Because of this finding and because the data set of Ref. (*2*) contains mostly duplicates to the ones of Ref. (*3*) we excluded the former from our fit. In total 539 transitions were fitted.

First we seek to show clearly the problem encountered by Maki *et al.* (*3*), when they tried to determine the potential energy in the Dunham form. We repeated the conventional Dunham fit as it is given in their Table IV because they partially imposed interrelations on U's. We sought to avoid such constraints in the first step to allow for all possible forms of an effective Hamiltonian, the energy levels of which are given by Eq. (6). This fit required 21 parameters, less than Maki *et al.* (*3*) had introduced, but attaining the same quality of the fit.

With freely floating U parameters and appropriate Δ one finds directly significant deviations between fitted U and predicted ones according to the interrelations of Dunham's theory; for instance, the predicted U_{02} deviates by 10 times the estimated standard deviation and U_{13} by 56 times. What do these deviations signify?

Because all effective Hamiltonians described above lead to the representation of the eigen energies in the same form of Eq. (6), the values of the "true" U are expected to obey the interrelations between each other and the corrections should be totally absorbed in the Δ. Thus the reason for the remaining deviations might be one or several of the following:

1. The molecule does not follow one of the above noted effective Hamiltonians in Eq. (4); perturbations of higher order should be applied to evaluate the electronic–nuclear interaction, or perturbation theory for deriving the effective Hamiltonian is inadequate for the case under consideration.
2. The perturbation approach to derive Eq. (6) from the effective Hamiltonian is inadequate.
3. The selection of the set of Dunham parameters and Δ is not the most favorable one; there might exist another set that describes the eigenenergies equally well. This fact will be one of our cautionary results in the present paper on LiH.

There exists no systematic way to select the set of parameters; therefore, for economic reasons the investigator normally will terminate the search when he finds a set which describes the measurements within given limits of error despite lack of conformity to the interrelations.

The success of Ogilvie (*4*) in finding a satisfactory fit to the data of LiH by the algebraic approach (i.e., the radial functions in the Hamiltonian are represented by power expansions) with simplification 2 proves that there exists a representation according to Eq. (6) that includes all interrelations of the coefficients U_{kl} up to the highest exponents of $(v + \frac{1}{2})$ and $J(J + 1)$ that the fitted potential-energy coefficients of $V(R)$ would allow to calculate. This form of truncation was invoked for the fit in Ref. (*4*).

To search for an appropriate U_{kl} and Δ_{kl} representation we applied the iterative fit procedure. Fits of the new type converged rapidly, typically with 6 to 10 iterations; the total duration of computation for the data set of LiH is only 15 sec by a microcomputer equipped with a CPU 80486 at 33 MHz. Seventeen parameters—appreciably less than the number stated above because we invoked the interrelations—were introduced to obtain a normalized standard deviation 1.06; this value demonstrates that the average deviations of the observed line positions are comparable the assigned uncertainties of the measurements. All the discrepancies encountered by Maki *et al.* (*3*) are removed; the Dunham formalism works also for LiH.

Ogilvie (*4*) obtained in his fit with exactly the same set of data and the same assumed experimental accuracies the same normalized standard deviation. Coxon (*5*) stated for the fit in the z representation (compare Eq. (9) below) a standard deviation of

0.951, which one cannot take as an indication that his fit is better than the present one because he mentioned that some lines for which the deviations were somewhat out of the expected range were removed from the fit. To such changes the standard deviation of a fit is most sensitive. The experimental errors assigned in our fit are the lower limits given in the original paper (*3*) and some will actually be higher than those used. Therefore, our fit result with $\sigma = 1.06$ is very satisfactory.

The results appear in Table I, which shows besides the values of the fitted $U_{kl}^{(1)}$, the calculated $U_{kl}^{(1)}$ for $l > 1$ and the corrections $U_{kl}^{(2)}$ which are necessary according to the interrelations from the Dunham potential, and the Δ_{kl} which contain only the corrections of the Born-Oppenheimer approximation. The potential coefficients c_j for an expansion with the normalized internuclear distance

$$z = 2(R - R_e)/(R + R_e), \qquad V(R) = c_0 z^2 (1 + \sum_j c_j z^j) \tag{9}$$

instead of x by Eq. (2) are calculated to simplify a direct comparison with existing fits from other sources, but the reader should be aware of the severe truncation problem discussed below.

Six parameters Δ for the hydrogen atom and three for the lithium atom were introduced in this fit. These are related to the functions α, β, and V_{cor}, but the expressions derived by Fernandez and Ogilvie (*12*) corresponding to these Δ contain more expansion coefficients than the number of Δ derived in the fit. This condition relates to the principal question whether adiabatic and nonadiabatic effects can be separated from spectral data obtained with field-free samples.

According to Ref. (*12*) not more than two of these functions can be determined from experimental observations, normally α and V_{cor} are chosen to remove the complication with the kinetic energy in Eq. (4); but then α and V_{cor} obtain a new definition because they absorb the effects of β somehow. Even with this simplification 2, the six Δ for H are given by seven expansion coefficients and the three Δ for Li by four. This indeterminacy cannot be removed by further measurements of the same type because any additionally determined Δ adds at least one coefficient of $\alpha(R)$ and/or $V_{cor}(R)$ to be derived. Thus splitting the Δ into separate contributions Δ^V and Δ^α solely from V_{cor} or α in the effective Hamiltonian leads to ambiguity in the case of pure rovibrational transitions. This might be overcome by incorporating experimental data of other types; we return to this point in the following section.

The separation of α and V_{cor} in Refs. (*4*, *5*) was based completely on the specific data set in hand and on the accidental preference which the nonlinear least squares fit gives to the choice of expansion coefficients. The fact that in both cases the value of three derived parameters is close to three or four times their estimated standard errors indicates that there may exist other combinations of parameters which fit the data equally well. Such poorly determined parameters do not appear for the results in Table I; in the worst case, e.g., Δ_{02}^{Li} has a value larger than five times the standard error, all others have an accuracy better than 5%.

On the other hand, the mass-independent potential energy $V(R)$ is extracted from the data by the mass scaling $1/\mu$ to the power $(k/2 + l)$ for $U_{kl}^{(1)}$ and by imposing interrelations between the coefficients U; the interrelations are derived from the Hamiltonian of Eq. (1). Both conditions define entirely unambiguously what is meant by the mechanical potential. Thus this part of the Hamiltonian is independent of the model applied in the analysis of the data. But some model dependence will always

TABLE I

Coefficients $U_{kl}^{(1)}$, Correction Terms $U_{kl}^{(2)}$, and Standard Errors σ_{kl} of LiH. The Column Labeled A Gives the Exponent for the Unit with Wavenumbers in cm^{-1} and the Column Labeled B Gives That in SI Units with Wavenumbers in m^{-1}.

k	l	$U_{kl}^{(1)}/cm^{-1}u^{(k/2+1)}$ A		$U_{kl}^{(2)}/cm^{-1}u^{(k/2+1+1)}$	$\sigma_{kl}/cm^{-1}u^{(k/2+1)}$	B
1	0	$/10^{0}$	1319.94948	-0.03496	0.0012	$/10^{2}$
2	0	$/10^{0}$	20.433289	0.000308	0.00095	$/10^{2}$
3	0	$/10^{0}$	0.146428		0.00035	$/10^{2}$
4	0	$/10^{-2}$	-0.23714		0.0042	$/10^{0}$
0	1	$/10^{0}$	6.62712470	-0.00002087	0.0000041	$/10^{2}$
1	1	$/10^{0}$	-0.17912971	-0.00000961	0.0000025	$/10^{2}$
2	1	$/10^{-2}$	0.161285		0.00018	$/10^{0}$
3	1	$/10^{-5}$	-3.1061		0.035	$/10^{-3}$
0	2	$/10^{-4}$	-6.6822205	-0.0006067		$/10^{-2}$
1	2	$/10^{-5}$	1.160655	0.000576		$/10^{-3}$
2	2	$/10^{-8}$	-8.69380			$/10^{-6}$
3	2	$/10^{-9}$	-4.06854			$/10^{-7}$
0	3	$/10^{-8}$	7.42994	0.00180		$/10^{-6}$
1	3	$/10^{-10}$	-4.03295			$/10^{-8}$
2	3	$/10^{-11}$	-5.50727			$/10^{-9}$
0	4	$/10^{-11}$	-1.07323	-0.00046		$/10^{-9}$
1	4	$/10^{-13}$	-2.05595			$/10^{-11}$
2	4	$/10^{-14}$	1.72947			$/10^{-12}$
0	5	$/10^{-15}$	1.81545			$/10^{-13}$
1	5	$/10^{-17}$	7.80158			$/10^{-15}$
0	6	$/10^{-19}$	-3.45347			$/10^{-17}$
1	6	$/10^{-20}$	-2.32051			$/10^{-18}$
0	7	$/10^{-23}$	7.03775			$/10^{-21}$
0	8	$/10^{-26}$	-1.50762			$/10^{-24}$

remain indirectly through the correlations with the model-dependent parameters such as expansion coefficients α_j and V_j introduced by the limited data set.

Figure 1 shows the difference between the potential in the z expansion obtained in this work and the potential determined by Coxon (*5*). The interval R used in Fig. 1 is calculated from the turning points of the highest vibrational level $v = 4$ included in the fit. Around the R_e of 1.6 Å the agreement between the potentials is very good as expected from the quality of the fits in both calculations and from the quantum mechanical foundations of both methods; but at the edge for large internuclear distances the divergence is quite obvious.

TABLE I—*Continued*

Coefficients Δ_{kl}^{Li} and Δ_{kl}^{H} and their standard errors for LiH X $^1\Sigma^+$

k	l	Δ_{kl}^{Li}	σ_{kl}^{Li}	Δ_{kl}^{H}	σ_{kl}^{H}
1	0	-0.12815	0.0056	-0.673763	0.00086
2	0			-0.4666	0.015
0	1	-0.15996	0.0068	-1.557598	0.00048
1	1			-0.8678	0.017
0	2	-0.595	0.11	-4.0798	0.030
0	3			-7.331	0.26

Coefficients c_j of the function V(z) for potential energy

j	c_j [a]	SI units
0	65724.8315 cm^{-1}	or 6572483.15 m^{-1}
1	-0.89727054	
2	0.34811778	
3	-0.08644651	
4	-0.04782896	
5	-0.02463798	
6	0.09516176	
R_e	1.5949085 Å	or 0.15949085 nm

a) No errors are shown for the c_j because the parameters U_{1k} should be used for predicting new molecular spectra and the presented c_j are only the first members of the series, which are calculable independently from each other applying the U_{kl}. Digits shown do not indicate the accuracy.

We should consider the problem of truncation in more detail. In principle the fit calculates the molecular parameters $U_{kl}^{(1)}$ and Δ_{kl}. Is there a unique way to derive the potential function $V(R)$ from the U's? The number of freely varied U will determine the number of independent potential coefficients, say c_0 to c_6 and R_e in the case of the eight U's fitted for LiH and they are shown in Table I; but do these parameters specify the function $V(R)$ completely? One obvious ambiguity becomes clear, if one calculates the a_j parameters according to an expansion with Eq. (2). There is a unique correspondence between the two sets a_j and c_j:

$$c_j = (j+1)! \sum_{k=0}^{j-1} a_{j-k}/[k!(j+1-k)!2^k] + (j+1)/2^j.$$

The same number of coefficients c or a is calculated from the derived $U_{kl}^{(1)}$ but the two potentials differ significantly because the coordinates applied in the expansions

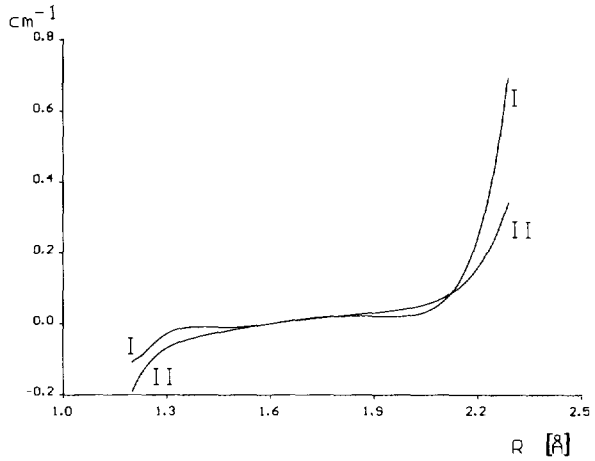


FIG. 1. Difference of the potential functions $V(R)$ for LiH determined by different methods: (I) difference between the potential of Coxon (*5*) and that from Table I; (II) difference between result from Table II and that from Table I. R is the conventional internuclear distance.

have a nonlinear relation between each other. This is shown in Fig. 2 for the case of LiH, the solid line is the difference of the two representations with c_j or a_j. Of course the main difference lies at the edge of the validity interval in R space, which should be roughly given by the classical turning points of the highest energy level contained in the fit, which is $v = 4$ in the present case. The full problem of specifying the potential comes out if one reminds the truncation of the energy formula (6); all U_{kl} higher than used in the fit were set to zero and they will determine mathematically an infinite number of potential coefficients, which are now all functions of and only of the few potential coefficients given in Table I. This leads directly to the convergence discussion of all power expansions, especially for the x expansion with the small convergence

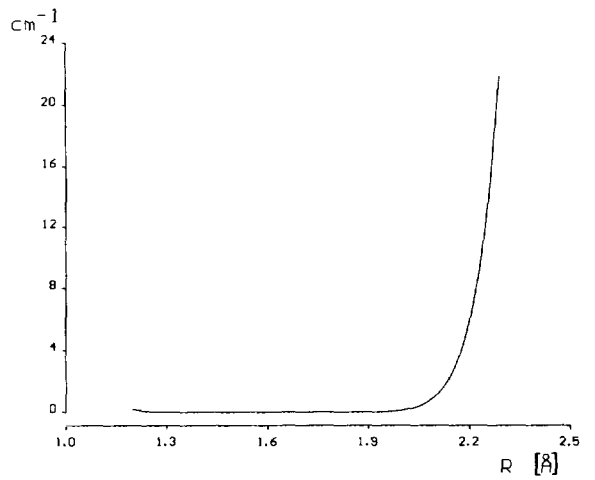


FIG. 2. Difference of the potential function $V(R)$ in the z (Eq. (9)) and the x (Eq. (2)) expansion as determined by the iterative fit procedure. R is the conventional internuclear distance.

radius already mentioned by other authors (*15*) and which was one of the reasons to introduce the expansion with z, Eq. (9).

We will note one other complication in calculating additional coefficients a_j or c_j from constrained parameters U_{kl}. If one uses the series U_{k0} and U_{k1} to derive more a_j or c_j the U_{kl} with $l > 1$, necessarily calculated from the interrelations, do not become zero as they should be from the truncation. From the practical point of view this will not be a serious problem because the high order terms U_{kl} will have little feedback to the observed energy levels and thus the resulting nonzero values will be consistent with the observations. We checked this by extending the potential to c_7 and got very satisfactory results.

From the mathematical point of view there are too many equations because for example each series of pairs $U_{kl}^{(1)}$ and $U_{kl'}^{(1)}$, with $l' - l = 1$ and $l \leqslant j_{\max} + 2$ will determine the full manifold of a_j or c_j extending the already known coefficients in the interval $0 \leqslant j \leqslant j_{\max}$.

To get physical consistency one can look for a truncation according to perturbation order. As shown by several authors (e.g., Ref. (*12*)) the perturbation order of Eqs. (1) and (3) is given by the exponent of the ratio $U_{01}^{(1)}/U_{10}^{(1)}$ and one can derive from Ref. (*8*) or Ref. (*12*)

$$U_{kl}^{(1)} \sim \left(\frac{U_{01}^{(1)}}{U_{10}^{(1)}}\right)^{k+2(l-1)} U_{01}^{(1)}$$

Therefore, if $a_{j\max}$ from the fit is determined by $U_{k\max,0}^{(1)}$ with $j_{\max} = 2k_{\max} - 2$, then the highest order involved is $k_{\max} - 1$ from $U_{k\max-1,1}^{(1)}$ and the set of a_j should be extended to include $U_{k\max+1,0}^{(1)}$ with this same order of perturbation. But this has the disadvantage that by the necessity to use $U_{k\max,1}^{(1)} = 0$ for deriving potential coefficients the order of perturbation is further increased. In total, the criterion of perturbation order is not consistently applicable and therefore would more likely direct us not to extend the set of potential coefficients by applying the $U_{k0}^{(1)}$ or $U_{k1}^{(1)}$ which were set to zero. But then the range of validity of the potential might be very limited in the internuclear distance space. This is a clear drawback of the algebraic approach compared to the method used by Coxon (*5*) because his derived potential is simply a finite polynomial and not a part of a power expansion as for the algebraic approach.

The present method has advantages in relation to the adiabatic and nonadiabatic corrections to which we will turn now.

SEPARATION OF ADIABATIC AND NONADIABATIC CONTRIBUTIONS

The nonadiabatic corrections originate from couplings of the electronic state under consideration with other electronic states. The molecular magnetic dipole moment of a state with $\Omega = 0$ is also a measure of these interactions; this quantity is known to give reliable estimates of the rotational nonadiabatic coupling (*16, 18*). The magnetic moment (conventionally represented by the rotational g-factor g_J) is measured from the Zeeman splittings of rotational transitions. As precise data have been derived from experiments with molecular beams on LiH (*17*), they should be applied to the analysis of the vibration–rotational spectra of LiH.

The observed rotational g-factor is related to the expectation value of $\alpha(R)$ of Eq. (5) for the specific vibrational state,

$$g_J = m_p(\langle \alpha^A \rangle / M_A + \langle \alpha^B \rangle / M_B), \tag{10}$$

with m_p as the proton mass. Such partition of g_J into isotopically independent parts was first used by Watson (*18*). For a diatomic molecule of polarity $^+AB^-$ but without net charge, the g-factor of the rotating molecule contains two contributions, one from the electronic–nuclear interaction and the other from the rotating electric dipole moment μ_e (*16, 18*),

$$g_J = g_J^{\text{nonad}} + m_p \mu_e / eR_e (1/M_A - 1/M_B) \tag{11}$$

in which e is the elementary electric charge. Being induced by the rotation, g_J^{nonad} is inversely proportional to the reduced mass of the molecule. Thus the expectation values in Eq. (10) become expressed as

$$\begin{aligned} \langle \alpha^A \rangle &= \mu g_J^{\text{nonad}} / m_p + \mu_e / eR_e \\ \langle \alpha^B \rangle &= \mu g_J^{\text{nonad}} / m_p - \mu_e / eR_e. \end{aligned} \tag{12}$$

Unfortunately Eq. (4) does not require the expectation values of α but its function of internuclear distance. If we assume α to be a slowly varying function with R, we can approximate the zeroth order expansion coefficient of α at the equilibrium internuclear distance to be the expectation value of a low vibrational state; we denote this coefficient by t_0^A or t_0^B: $t_0^A \approx \langle \alpha^A \rangle$ and $t_0^B \approx \langle \alpha^B \rangle$.

Because we use for this interpretation z from Eq. (9) as expansion parameter the coefficients similar to Eq. (2′) are renamed to:

$$\text{for } V_{\text{cor}}(R) \rightarrow u_j, \qquad \text{for } \alpha(R) \rightarrow t_j, \qquad \text{and for } \beta(R) \rightarrow s_j.$$

Drawing the g-factors from Ref. (*17*) and the dipole moment from Ref. (*19*), we find

$$t_0^{\text{H}} = -0.7682 \qquad \text{and} \qquad t_0^{\text{Li}} = +0.7534. \tag{13}$$

This information we can directly relate to the value of Δ_{01} given in Table I, because the present analysis does not distinguish between the different simplifications of the Hamiltonian in Eq. (4). According to Fernandez and Ogilvie (*12*), we have the relation

$$\Delta_{01} = t_0 + 4U_{01}/U_{10}^2 u_1, \tag{14}$$

in which u_1 is the linear expansion coefficient of $V_{\text{cor}}(z)$.

Because of the low accuracy of the parameter Δ and the approximations already made it is not necessary to distinguish in Eq. (14) U_{kl} from $U_{kl}^{(1)}$. By means of U_{01} and U_{10} from Table I we derive $u_1^{\text{H}} = -5.19 \times 10^4\ \text{cm}^{-1}$ and $u_1^{\text{Li}} = -6.00 \times 10^4\ \text{cm}^{-1}$; we estimate the uncertainties to be about 1% for H and 3% for Li taking into account the accuracy of Δ_{01}, of g_J, the dipole moment, and most critically the approximation made for t_0.

One detects significant differences up to a factor of 2 on u_1 of H and even 7 of Li between our present result and the previous ones (*4, 5*). This effect is a clear indication of the missing part given by the function β in the effective Hamiltonian or of the neglected first expansion coefficient t_0 in Refs. (*4, 5*). Our fit uses the full Hamiltonian, the derived parameters Δ_{lk} contain the appropriate linear combinations of the expansion coefficients of V_{cor}, α, and β. To confirm this deduction we use the following estimation of the first expansion coefficient of β.

If one calculates the nonadiabatic part of the g-factor for LiH (comp. Eq. (11)) this comes out to be almost negligible, for ^{7}LiH $g_J^{\text{nonad}} = -0.008(2)$, which means the rotation correction through α is given by the rotating dipole moment, i.e., by the

electron mass shift within the ionic molecule in the lowest order. The functional form of the Hamiltonian will change in the case of the lowest order in α and β to a Hamiltonian with the effective mass

$$\mu = \frac{(M_A - m_e d)(M_B + m_e d)}{M_A + M_B} \quad \text{yielding } \frac{1}{\mu} = \frac{1}{\mu_0}\left(1 + \frac{m_e d}{M_A} - \frac{m_e d}{M_B} + \cdots\right), \quad (15)$$

where d is the part of the electron shifted according to the dipole moment, e.g., $d = \mu_e / eR_e$ and μ_0 is the reduced molecular mass in the unperturbed case. Such mass corrections for ionic molecules were already applied, for example by Ryzlewicz *et al.* (*20*) for BaCl.

If we introduce this effective mass for the full Hamiltonian in Eq. (4) by using the above approximation for $1/\mu$ then the lowest order coefficient of β becomes $s_0^A = +d$ and $s_0^B = -d$, which results for LiH to $s_0^{Li} = -s_0^H = +0.7608$ and the same for α. With these estimates a fit of all LiH data was performed similarly to that in Ref. (*4*) but applying now the full algebraic model according to Ref. (*12*), the result of which is

TABLE II

Coefficients of the Radial Functions of LiH $X\,^1\Sigma^+$ from a Nonlinear Least-Squares Fit with the Algebraic Method

j	c_j	t_j^{Li}	t_j^H	$u_j^{Li}/10^4 cm^{-1}$ /$10^6 m^{-1}$	$u_j^H/10^4 cm^{-1}$ /$10^6 m^{-1}$
0	(65724.917 ±0.075)cm^{-1}	[0.75335]	[-0.76823]	...	...
	(6572491.7±7.5)m^{-1}				
1	-0.8972601 ±0.0000135	0.512 ±0.111	0.863 ±0.029	-5.683 ±0.046	-5.1892 ±0.0035
2	0.347982 ±0.000095		-1.178 ±0.103	6.976 ±0.130	8.3021 ±0.0121
3	-0.08666 ±0.00042				-10.83 ±0.26
4	-0.04394 ±0.0022				9.31 ±0.84
5	-0.0308 ±0.0058				
6	0.0758 ±0.0109				
	[s_0^{Li} = 0.7608]				
	[s_0^H = -0.7608]				

$U_{10}^{(1)}$ = (1319.94784±0.00117) cm^{-1} $u^{1/2}$

(131994.784±0.117) m^{-1} $u^{1/2}$

$U_{01}^{(1)}$ = (6.6270996±0.0000042) cm^{-1} u

(662.70996±0.00042) m^{-1} u

R_e = (1.59491127±0.00000143) Å

(0.159491127±0.000000143) nm

Parameters in brackets were constrained to the specified values; the normalized standard deviation of the fit was 1.10.

presented in Table II. The parameters are given in the z expansion according Eq. (9). The normalized standard deviation was 1.10 with 17 freely varied parameters and this is quite similar to the fit quality of the parameter set of Table I. The derived potential is compared with that calculated from the results in Table I and the corresponding difference is included in Fig. 1. The agreement is good, only slight deviations at the edges are recognized which might originate from the somehow artificial separation between V_{cor} and α for the higher orders.

Because we introduced for α and β the lowest order, the lowest order of the u parameters of V_{cor} should be interpretable containing only little contributions from neglects of higher order terms in β. The values of u_1 in Table II agree almost within error limits to the values given above from the analysis of the Δ parameters despite the approximations made.

This gives us further confidence to compare also the obtained t_1^H and t_1^{Li} in Table II with values calculable from the rotational dependence of the g-factor as reported in Ref. (*17*). From this data one estimates the linear expansion coefficient of the g factor to be 2.1(5) for the normalized coordinate z. Applying again Eqs. (11) and (12) now for the expansion coefficients neglecting the internuclear distance dependence of μ_e one gets for $t_1^H = +1$ and $t_1^{Li} = +2$ with an error of 50 to 100%. This is the right order of magnitude and the right sign and does not contradict the fitted values of Table II because one has to consider that only the lowest order in β was introduced.

One could perform similar interpretations starting solely from the Δ of Table I; the known values of t_0 and u_1 can be used in connection with the estimate of s_0 to calculate from Δ_{10} the value of u_2 and then from Δ_{02} the value of t_1. The formulas are given in Ref. (*12*). The result is comparable with the direct fit. But it is not possible to proceed further because of the lack of reasonable estimates of higher orders in the function β; the knowledge of higher orders in α, which could be obtained if the Zeeman effect is measured for different vibrational states, is insufficient. The higher order separations given in Table II as t_j and u_j are suspect because of lacking s_j and cannot be used for physical interpretation.

CONCLUSIONS

The three methods of fitting such precise data sets as that of LiH all show the internal consistency of the body of data. The molecular parameters obtained from the fits can be used to predict with great reliability the frequencies of unobserved vibration–rotational transitions at least within the same range of quantum numbers included in the fit. For instance by means of parameters in Table I pure rotational transitions of vibrational levels up to $v = 2$ can be predicted with a standard deviation less than 3 MHz for J below 20 increasing to 6 MHz for values up to 30. Little can be stated with certainty for predicting frequencies for higher J, because we would extrapolate outside the range of observations. A similar situation holds for the vibrational transitions.

From the present analysis we obtained no indication of any limitation of the second-order perturbation treatment for the corrections of the Born–Oppenheimer approximation performed in Ref. (*9*) to derive the effective Hamiltonian of Eq. (4). This question was raised by Maki *et al.* (*3*) when they did not succeed in fitting their observations. The set of molecular parameters which they chose was not adequate for the case of LiH. The isotope corrections for LiH could be traced back to main contributions by the rotating and vibrating large electric dipole moment, nonadiabatic effects are very small.

The present analysis gives reliable estimates of the mass independent Born–Oppenheimer potential and does not incorporate model dependence. The derived potentials in this study and by Coxon (*5*) do agree within the reduced range of validity which comes about by the truncation built into the algebraic approach, but the range of validity of the potential derived by Coxon might also be smaller than expected because by neglecting the function $\beta(R)$ in the Hamiltonian the potential can be influenced.

Coxon tried to use the molecular parameters determined by Ogilvie (*4*) for calculating rovibrational transitions with his method but could not reproduce the observations within acceptable limits. This is certainly related to the truncation, and a reliable estimate of the range of validity of the potential determined in the algebraic approach is needed. Furthermore, for a full application of the algebraic method a better understanding of the influence of the expansion coordinate must be explored (comp. Fig. 2). This would be to some extent a continuation of the investigation in Ref. (*15*), where the convergence of the different power expansions of the potential was discussed. Computational studies are under way to test different expansion coefficients and the results will be published (*21*) in connection with the computer program applied in the present work because of its effectiveness to analyze not only rovibrational spectra but also electronic transitions for molecules in Hund's coupling case c.

The derived radial functions α and V_{cor} as in Refs. (*4, 5*) must be taken with great care, because the effective Hamiltonian has to be complete in order to draw any physical conclusion from them. In this respect the fit with the Δ's and fully imposed interrelations between the U's does not suffer and give an easy first separation of the lowest orders in the adiabatic and nonadiabatic terms, if the molecular g-factor is known from other sources. The present analysis shows clearly that all other trials of separating these effects by the use of pure spectroscopic data from zero external field configurations lead to ambiguities. Only linear combinations as given by the Δ's for example are determinable and from these an infinite number of sets of correction functions α, β, and V_{cor} can be derived.

The numerical integration of the Schrödinger equation applied by Coxon (*5*) certainly has the advantage to put an analytic form of the potential into the calculation and avoiding the truncation. In this way the method extracts directly the Born–Oppenheimer potential but it suffers from the incomplete Hamiltonian. In his application to LiH, Coxon used more mass-dependent parameters than we introduced to get a satisfactory fit. It has to be studied if this has significant drawbacks to the determination of the potential itself because of correlations. For the representation of the potential with the variable-β Morse function, Coxon (*5*) introduces even more parameters, and it is not immediately obvious if this would lead to any reliable determination of the isotope effects. Any possible inappropriateness of the analytic form could be compensated by artificial radial functions of the Born–Oppenheimer corrections.

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